

Rehan Sanjay Venkatesan

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Profile

New Computer Science graduate building real-time voice AI agents. **12 fixes merged** into open-source voice-agent frameworks — **8 in LiveKit Agents**, including a core connection-pool fix that repaired 11 plugins at once. Built **Atlas**, an LLM phone agent, end to end on my own.

Experience

LiveKit Agents, Pipecat, jambonz, drachtio

Open-Source Contributor

Remote

Jul 2026–Present

- 12 PRs merged across four voice-AI frameworks; 8 into LiveKit Agents (14k stars), reviewed by its core team — two merged by the co-founder.
- Core framework fix (#7139): the shared connection pool closed sockets that were still streaming audio, and reused a connection opened with old settings. Fixed it in the pool itself, repairing all 11 plugins that depend on it.
- Found the same stale-connection bug in 3 TTS plugins and fixed each: asyncai #7132, neuphonic #7133, cartesio #7140.
- Removed a class of silently lost asyncio tasks: fixed 15 instances and got ruff's RUF006 rule enabled in LiveKit's CI so it cannot return (#7054).
- Fixed resources left open after shutdown — an STT connection pool, per-stream HTTP sessions and a prewarm task (#7023, #7012, #7050).
- Pipecat (#5464): runtime model, voice and language changes were stored but never sent to the provider. SIP / telephony: duplicate call-status webhooks after a transfer (jambonz #1584), resumed-TTS events sent to the wrong hook (jambonz #1585), wrong headers stripped from proxied responses (drachtio-srf #240).
- Every fix ships with a regression test. 7 more PRs under review; I also review other contributors' PRs.

Codtech IT Solutions

Backend Web Development Intern

Chennai

Feb–Mar 2025

Java and MySQL backend modules — CRUD, authentication, REST APIs, secure login.

Makos Infotech

Front End Development Intern

Chennai

Jun–Jul 2024

Responsive React, HTML5 and CSS3 components in an Agile team.

Projects

Atlas

LLM voice agent for phone calls — built solo, ~31,000 lines

2026

- Real-time pipeline: Twilio Media Streams → Deepgram streaming speech-to-text → LLM (Groq) → Deepgram text-to-speech, over WebSockets.
- Conversation handling: barge-in stops the agent when the caller talks; replies only after Deepgram signals end of speech; an instant "mm-hmm" masks LLM latency; a silence watchdog.
- After each call an LLM classifies the outcome and books callbacks, with WhatsApp follow-up; outbound dialling only within legal calling hours. Tested end to end on real calls.
- NestJS, Next.js, Postgres/Prisma, Redis/BullMQ. Completed Twilio business verification.

InvoiceCheck.in: Live GST invoice checker: 11-point compliance check before marketplaces reject an invoice, OCR auto-fill, Razorpay payments. Next.js, TypeScript, Supabase.

Coral Reef: CNN classifying coral bleaching from 5,000+ underwater images. Python, TensorFlow, Flask.

Skills

Languages: Python (asyncio, pytest), TypeScript, JavaScript, Java, SQL

Voice AI: LiveKit Agents, Pipecat, Deepgram, Twilio Media Streams, WebRTC, SIP (jambonz, drachtio)

LLMs: OpenAI, Anthropic and Groq APIs

Backend: NestJS, Node.js, Flask, Postgres, Prisma, Redis, BullMQ, Docker, CI/CD

Frontend: React, Next.js, Tailwind

Education

Sri Sairam Institute of Technology

B.E. Computer Science and Engineering, CGPA 7.24

Chennai

2022–2026